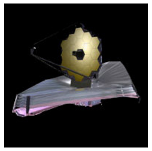

8330 - Calibration of the Brighter-Fatter Effect in AMI with a Reference Binary

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	2	EZ Aqr - AMI	NIRISS Aperture Masking Interferometry	(3) V-EZ-Aqr

ABSTRACT

High resolution, high contrast imaging with JWST is severely limited by the 'brighter-fatter effect', or charge migration, whereby bright pixels nonlinearly bleed charge into their neighbours. If this problem can be resolved, the door is open to much more extensive use of JWST for high contrast imaging. We propose using the Aperture Masking Interferometer on NIRISS as an ideal calibration system, as the non-redundant mask makes phase retrieval a well-posed problem and the PSF can be accurately characterized, so that systematic effects from the detector can be

separately determined. We propose training a differentiable optical model end-to-end with a neural network to represent this nonlinear detector, but existing data on point sources and flat fields do not contain enough PSF diversity to avoid over-fitting in training the neural network. We therefore request AMI and clear-pupil NIRISS observations of the nearby, well-characterized, moderately-bright M dwarf system EZ Aqr, as a 'ruler' on the sky whose precisely-known geometry will unambiguously constrain a detector model in conjunction with existing point source and flat field observations. With an accurate, data-driven model of the brighter-fatter effect in hand, it will be possible to recover strong science outcomes in exoplanet, disk, and AGN science from archival GO and GTO AMI data; plan future observations with this instrument; and pave the way for applications to enhancing coronagraphic NIRCam and Roman observations.

OBSERVING DESCRIPTION

We propose observing the nearby, near-equal binary EZ Aqr as a benchmark star for NIRISS/AMI calibration. We will observe this at 2 primary dithers by 5 subpixel dithers, in each of the available F380M, F430M, and F480M filters, with AMI and with the standard CLEARP imaging modes. We do not require a PSF reference star: this binary system itself will be used as a PSF reference.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	V-EZ-Aqr	RA: 22 38 33.5760 (339.6399000d) Dec: -15 17 59.76 (-15.29993d) Equinox: J2000	Proper Motion RA: 2314.8 mas/yr Proper Motion Dec: 2295.3 mas/yr Parallax: 0.2936" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Calibration
Description=[Binary stars, Point spread function]

Observation

Proposal 8330, Observation 2: EZ Aqr - AMI

Fri Jun 27 15:00:45 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRISS Aperture Masking Interferometry

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

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Category=Calibration
Description=[Binary stars, Point spread function]

Acquisition

#	Target	Acquisition Mode	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	3 V-EZ-Aqr	AMIBRIGHT	F480M	NISRAPID	9	1	1	0.475	227606.5

Template

Subarray
SUB80

Dithers

#	Primary Dithers	Subpixel Positions
1	2	5
2	NONE	NONE

Direct Image

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F430M	NISRAPID	3	455	1	455	146.619	
2	F380M	NISRAPID	3	500	1	500	161.12	
3	F480M	NISRAPID	3	325	1	325	104.728	

Proposal 8330 - Observation 2 - Calibration of the Brighter-Fatter Effect in AMI with a Reference Binary

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	Direct Image
1	F480M	NISRAPID	18	220	10	2200	3198.448	227606.3	false
2	F380M	NISRAPID	11	260	10	2600	2406.976	227606.9	false
3	F430M	NISRAPID	16	250	10	2500	3257.4	227606.10	false
4	F430M	NISRAPID	3	455	10	4550	1466.192	227606.32	true
5	F380M	NISRAPID	3	500	10	5000	1611.2	227606.31	true
6	F480M	NISRAPID	3	325	10	3250	1047.28	227606.22	true

PSF References

Additional Justification: true

Special Requirements

No Parallel Attachments